



Ext 2

NORTH SYDNEY BOYS HIGH SCHOOL

2011
ASSESSMENT TASK 2

Mathematics

Extension 2

General Instructions

- Working time – 55 minutes
- Write in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.

- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Fletcher
- ☐ Mr Ireland
- ☐ Mrs Collins/Mr Rezcallah

Student Number:

(To be used by the exam markers only.)

Question No	1	2	3	Total	Total
Mark	15	15	8	38	100

Question 1 (15 marks)

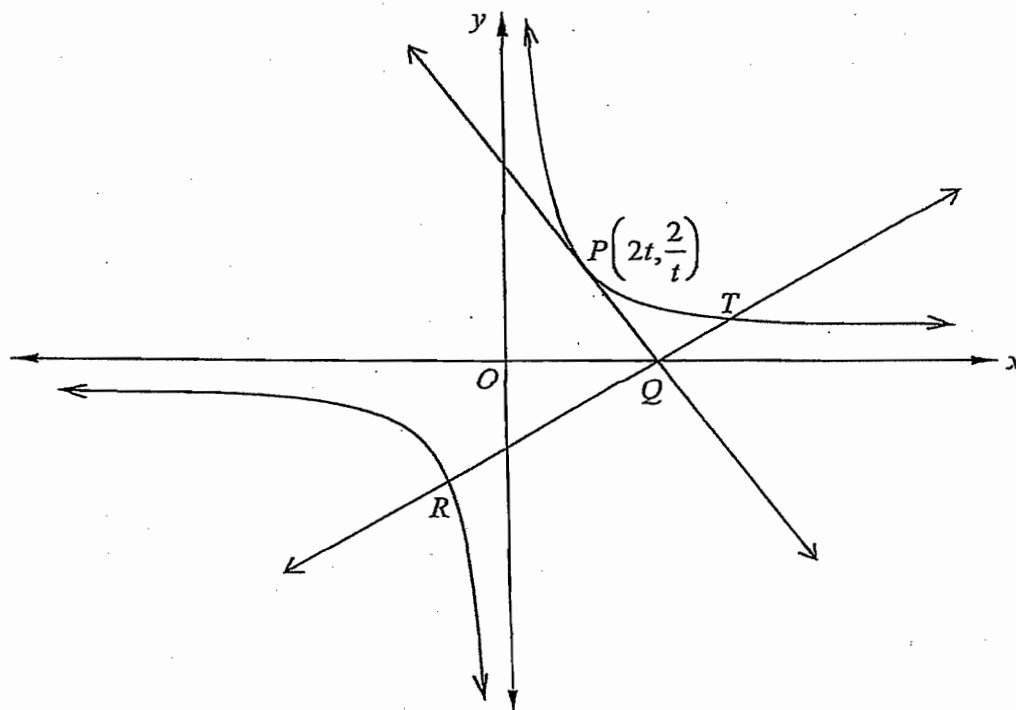
Marks

(a) For the hyperbola $\frac{x^2}{144} - \frac{y^2}{25} = 1$ find

- (i) The eccentricity. 1
- (ii) The co-ordinates of the foci. 1
- (iii) The equations of the asymptotes. 1
- (iv) The equations of the directrices. 1

Sketch the graph of the hyperbola showing the above information. 2

(b) $P\left(2t, \frac{2}{t}\right)$ is a point on the rectangular hyperbola $xy = 4$. The tangent at P cuts the x -axis at Q and the line through Q , perpendicular to the tangent at P , cuts the hyperbola at the points R and T as shown.



- (i) Show that the equation of the tangent at P is $x + t^2y = 4t$. 2
- (ii) Show that the line through Q , perpendicular to the tangent at P , has equation $t^2x - y = 4t^3$. 3
- (iii) If M is the midpoint of RT , show that M has coordinates $(2t, -2t^3)$. 3
- (iv) Find the equation of the locus of M , as P moves on the curve $xy = 4$. 1

Question 2 (15 marks)

Marks

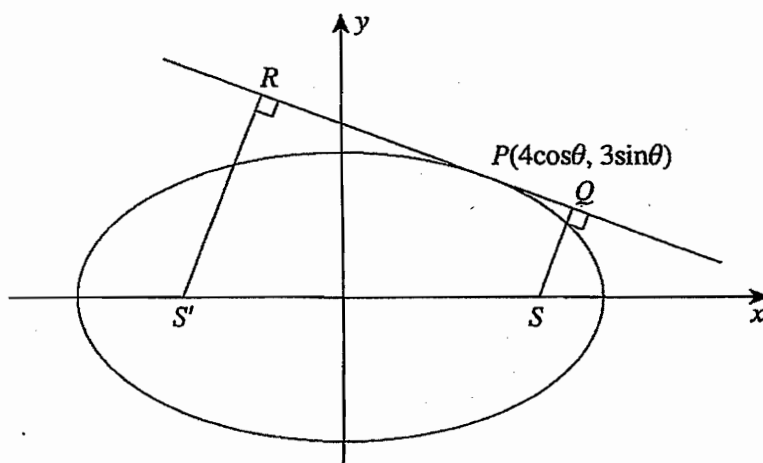
(a) Consider the ellipse $\frac{x^2}{16} + \frac{y^2}{12} = 1$

- (i) Use differentiation to derive the equations of the tangent and normal to the ellipse at the point $P(2, 3)$ 3
- (ii) The tangent and normal to the ellipse at P cut the y axis at A and B respectively. Find the co-ordinates of A and B . 2
- (iii) Show that AB subtends a right angle at the focus S of the ellipse 2
- (iv) Show that A, P, S and B are concyclic 1
- (v) Find the centre and radius of the circle which passes through the points A, P, S and B . 2

(b) Consider the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

A tangent to this ellipse is drawn at point $P(4 \cos \theta, 3 \sin \theta)$

Perpendiculars are drawn from each focus of the ellipse to meet the tangent at Q and R as shown on the diagram.



- (i) Prove that the equation of the tangent at P is $\frac{x \cos \theta}{4} + \frac{y \sin \theta}{3} = 1$. 2
- (ii) Show that $QS \times RS' = 9$. 3

Question 3 (8 marks)**Marks**

(a) Resolve $\frac{1}{1-x^4}$ into partial fractions

3

(b) (i) Prove by Mathematical Induction that if n is a positive integer,
then $2^{(n+4)} > (n+4)^2$

4

(ii) By choosing a suitable substitution, or otherwise, show that
if a is a positive integer, then $2^{3(a+2)} > 9(a+2)^2$

1

SAMPLE ANSWERS

CRITERIA

MARKS

1 (a) $\frac{x^2}{144} - \frac{y^2}{25} = 1$ $a=12, b=5$

(i) $b^2 = a^2(e^2 - 1)$

$e^2 = \frac{25}{144} + 1$

$= \frac{169}{144}$

$e = \frac{13}{12}$

$e > 0$

Finds eccentricity

1

(ii) foci $(\pm ae, 0)$

$ae = 12 \times \frac{13}{12}$

$\therefore$ foci $(\pm 13, 0)$

Finds coords of foci

1

(iii) Asymptotes $y = \pm \frac{b}{a}x$

$y = \pm \frac{5}{12}x$

Finds equations of asymptotes

1

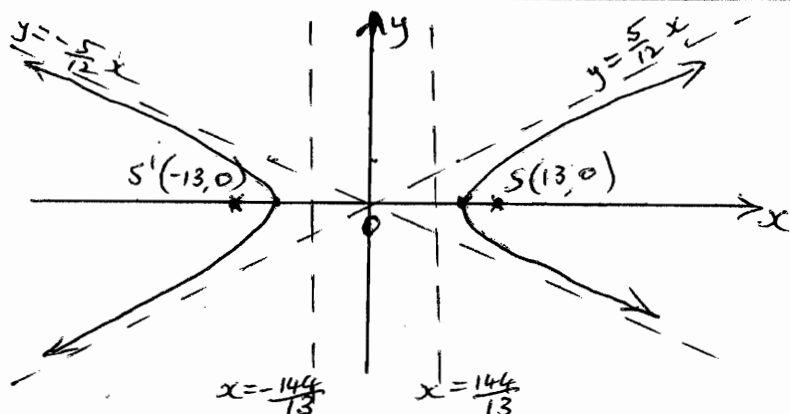
(iv) Directrices

$x = \pm \frac{a}{e}$

$x = \pm \frac{144}{13} (= \pm 11 \frac{1}{13})$

Finds equations of directrices

1



Graphs correct shape determined by correctly labelled asymptotes

1

Indicates correctly directrices & foci

1

(b) (i) $y = 4x^{-1}$ $\frac{dy}{dx} = -4x^{-2}$

At $x = 2t$, $m = -\frac{1}{t^2}$

$\therefore$ Equation of tangent is

$y - \frac{2}{t} = -\frac{1}{t^2}(x - 2t)$

$t^2y - 2t = -x + 2t$

$x + t^2y = 4t$

Finds correct gradient or makes significant progress towards solution.

1

Shows correct equation

1

(ii) At Q, $y = 0$ $\therefore$ from (i) $x = 4t$

i.e. Q $(4t, 0)$

Line through Q $\perp$ tangent at P has

$m = t^2$

$\therefore$ Equation of line is

$y - 0 = t^2(x - 4t)$

$y = t^2x - 4t^3$

$\therefore t^2x - y = 4t^3$

Finds correct coords of Q

1

Finds correct gradient or makes significant progress towards solution

1

Shows correct equation

1

(iii) R and T lie on the hyperbola and the line through Q. Solve $xy = 4$ and $t^2x - y = 4t^3$

$x(t^2x - 4t^3) = 4$

$t^2x^2 - 4t^3x - 4 = 0$ Let roots be α, β .

then $\alpha + \beta = \frac{4t^3}{t^2} = 4t$

M(x, y) is halfway between the roots

$\therefore x = \frac{\alpha + \beta}{2} = 2t$

As y lies on RT $y = t^2(2t) - 4t^3 = -2t^3$

$\therefore M(2t, -2t^3)$

Establishes correct equation

1

Finds x or y coordinate

1

Finds the midpoint

1

SAMPLE ANSWERS

CRITERIA

MARKS

① (iv) At M , $x = 2t$ $y = -2t^3$
 $\therefore y = -2\left(\frac{x}{2}\right)^3 = -\frac{x^3}{4}$

Thus the locus of M is the curve $y = -\frac{x^3}{4}$, $x \neq 0$, $y \neq 0$

Finds the locus

1

② (i) $\frac{x^2}{16} + \frac{y^2}{12} = 1 \Rightarrow \frac{2x}{16} + \frac{2y}{12} \frac{dy}{dx} = 0$
 (a) $\frac{dy}{dx} = -\frac{x}{8} \div \frac{y}{6} = -\frac{3x}{4y}$

At $P(2,3)$ $m = -\frac{1}{2}$

$\therefore$ Equation of tangent at P is:

$$y - 3 = -\frac{1}{2}(x - 2)$$

$$2y - 6 = -x + 2$$

$$x + 2y - 8 = 0$$

Equation of normal at P is

$$y - 3 = 2x - 4$$

$$2x - y - 1 = 0$$

Obtains correct expression for $\frac{dy}{dx}$

1

Finds correct equation of tangent

1

Finds correct equation of normal

1

(ii) Sub $x = 0$ into equation of tangent
 $2y - 8 = 0 \Rightarrow A(0, 4)$

Sub $x = 0$ into equation of normal
 $-y - 1 = 0 \Rightarrow B(0, -1)$

Finds coords of A

1

Finds coords of B

1

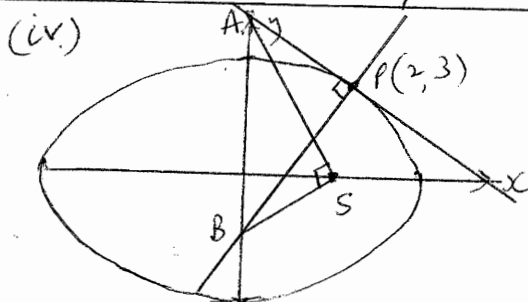
(iii) $b^2 = a^2(1 - e^2) \Rightarrow e = \frac{1}{2}$
 Foci $(\pm ae, 0) \Rightarrow S(\frac{1}{2} \times 4, 0) = (2, 0)$
 $S(2, 0)$ $A(0, 4)$ $B(0, -1)$
 $m_{AS} \cdot m_{BS} = -2 \times \frac{1}{2}$
 $= -1$
 $\therefore \angle ASB = 90^\circ$

Correctly finds coords of focus

1

Shows $m_{AS} \cdot m_{BS} = -1$

1



AP is tangent at P
 BP is normal at P
 $\therefore \angle APB = 90^\circ$

$\therefore AB$ subtends equal angles of 90° at P and S

$\therefore APSB$ are concyclic

Shows A, P, S and B are concyclic.

1

(v) Diameter is AB , so centre is $(0, \frac{3}{2})$
 Radius is $2\frac{1}{2} = \frac{5}{2}$

Finds centre

1

Finds radius

1

Sample Answers

Criteria

Marks

(2)(b)(i) $x = 4 \cos \theta$ $y = 3 \sin \theta$
 $\frac{dx}{d\theta} = -4 \sin \theta$ $\frac{dy}{d\theta} = 3 \cos \theta$
 $\frac{dy}{dx} = -\frac{3 \cos \theta}{4 \sin \theta}$

Equation of tangent at P is:

$$y - 3 \sin \theta = -\frac{3 \cos \theta}{4 \sin \theta} (x - 4 \cos \theta)$$

$$4y \sin \theta - 12 \sin^2 \theta = -3x \cos \theta + 12 \cos^2 \theta$$

$$3x \cos \theta + 4y \sin \theta = 12(\sin^2 \theta + \cos^2 \theta)$$

$$\frac{3x \cos \theta}{12} + \frac{4y \sin \theta}{12} = 1$$

$$\frac{x \cos \theta}{4} + \frac{y \sin \theta}{3} = 1 \text{ as required.}$$

Correctly finds $\frac{dy}{dx}$

1

Uses correct process to find required equation.

1

(ii) $a = 4$, $b = 3$ $c = \sqrt{1 - \frac{9}{16}} = \frac{\sqrt{7}}{4}$

$\therefore S$ and S' have coords $(5\sqrt{7}, 0)$ and $(-5\sqrt{7}, 0)$ respectively

Equation of tangent is

$$3x \cos \theta + 4y \sin \theta - 12 = 0$$

So, $d_{QS} = \frac{|3\sqrt{7} \cos \theta - 12|}{\sqrt{9 \cos^2 \theta + 16 \sin^2 \theta}}$

$$d_{RS'} = \frac{|-(3\sqrt{7} \cos \theta + 12)|}{\sqrt{9 \cos^2 \theta + 16 \sin^2 \theta}}$$

$$\begin{aligned} \therefore QS \times RS' &= \frac{|-(63 \cos^2 \theta - 144)|}{9 \cos^2 \theta + 16 \sin^2 \theta} \\ &= \frac{|9(16 - 7 \cos^2 \theta)|}{9 \cos^2 \theta + 16(1 - \cos^2 \theta)} \\ &= \frac{|9(16 - 7 \cos^2 \theta)|}{16 - 7 \cos^2 \theta} \\ &= 9 \text{ as required.} \end{aligned}$$

Correctly finds c and coords of S and S'

1

Correctly finds expressions for d_{QS} and $d_{RS'}$

1

Simplifies the product of d_{QS} and $d_{RS'}$ correctly to give the desired result.

1

SAMPLE ANSWERS	CRITERIA	MARKS
(3) (a) $\frac{1}{1-x^4} = \frac{A}{1-x} + \frac{B}{1+x} + \frac{Cx+D}{1+x^2}$ $= \frac{A(1+x)(1+x^2) + B(1-x)(1+x^2) + (Cx+D)(1-x^2)}{1-x^4}$ $= \frac{A(1+x+x^2+x^3) + B(1-x-x^2-x^3) + Cx+D-Cx^3-Dx^2}{1-x^4}$ $= \frac{(A-B-C)x^3 + (A+B-D)x^2 + (A-B+C)x + (A+B+D)}{1-x^4}$ $A-B-C=0$ — (1) $A+B-D=0$ — (2) $A-B+C=0$ — (3) $A+B+D=1$ — (4) $(4)-(2) \Rightarrow 2D=1$ $D=\frac{1}{2}$ $(3)-(1) \Rightarrow 2C=0$ $C=0$ $\therefore A-B=0$ — (5) $A+B=\frac{1}{2}$ — (6) $\Rightarrow A=\frac{1}{4}, B=\frac{1}{4}$ $\therefore \frac{1}{1-x^4} = \frac{1}{4(1-x)} + \frac{1}{4(1+x)} + \frac{1}{2(1+x^2)}$	correctly factorises $1-x^4$ and forms sum of 3 algebraic fractions Correctly forms equations Finds correct values for A, B, C, D $\frac{1}{1-x^4} = \frac{1}{2(1-x^2)} + \frac{1}{2(1+x^2)}$	1 1 1 Max 2
(b) (i) RTP $2^{(n+4)} > (n+4)^2$ For $n=1$, $2^5 > 5^2$ $32 > 25$ which is true Assume the result is true for $n=k$ i.e. $2^{(k+4)} > (k+4)^2$ Consider $n=k+1$, then $2^{k+1+4} = 2(2^{k+4})$ i.e. $2^{k+5} > 2(k+4)^2$ RTP $2^{k+5} > 2(k+4)^2$ $= 2k^2 + 16k + 32$ $= k^2 + 10k + 25 + k^2 + 6k + 7$ $= (k+5)^2 + k^2 + 6k + 7$ Now, since k is a positive integer, $k^2 + 6k + 7 > 0$ $\therefore 2^{k+5} > (k+5)^2$ i.e. $2^{[(k+1)+4]} > [(k+1)+4]^2$ $\therefore$ If the result is true for $n=k$, it is also true for $n=k+1$ Now the result is true for $n=1$, hence it is true for $n=2$ and $n=3$ and so on for all positive integers n.	Correctly shows result is true for $n=1$ uses correct process to arrive at $2^{k+5} > (k+5)^2$ Shows $k^2 + 6k + 7 > 0$ since $k > 0$ Shows $2k^2 + 16k + 32 > k^2 + 10k + 25$ Correctly shows result is true for $n=k+1$, given it is true for $n=k$ and concludes by induction result is true for all n i.e. must show true for $n=1, n=2, \dots$	1 1 1
(ii) Now let $3(a+2) = n+4$ $\therefore 3a+2 = n$ Sub. $n=3a+2$ in $2^{n+4} > (n+4)^2$ $2^{3a+2+4} > (3a+2+4)^2$ $2^{3a+6} > [3(a+2)]^2$ $\therefore 2^{3(a+2)} > 9(a+2)^2$	uses substitution $n=3a+2$ to arrive at required result Numerical substitution Correct Proof by Mathematical Induction	1 0 1